Zigzags and Triangles

Fall 2021 ARML Power Contest

<u>The Solutions</u>

1. (a) The corresponding triangular array begins

$$\begin{array}{ccccccccccc}
&&&&&1&&&&& \\
&&&&1&&2&&&& \\
&&&1&&2&&4&&& \\
&&1&&2&&4&&8&& \\
&1&&2&&4&&8&&16& \\
1&&2&&4&&8&&16&&32
\end{array}$$

thus the triangular transform of the sequence—the terms along the right-hand edge of this triangle—is 1, 2, 4, 8, 16, 32, ...

(b) The corresponding triangular array begins

$$\begin{array}{ccccccccccc}
&&&&&1&&&&& \\
&&&&-1&&0&&&& \\
&&&1&&0&&0&&& \\
&&-1&&0&&0&&0&& \\
&1&&0&&0&&0&&0& \\
-1&&0&&0&&0&&0&&0
\end{array}$$

thus the triangular transform of the sequence—the terms along the right-hand edge of this triangle—is 1, 0, 0, 0, 0, 0, ...

2. Reverse the process of creating a corresponding triangular array by placing the given sequence on the right-hand edge of the triangle and subtracting the number above left of the current term to find the next term to the left. Then, the sequence that appears on the left-hand edge of this triangle will be a sequence that would transform into the given sequence.

$$\begin{array}{ccccccccccc}
&&&&&0&&&&& \\
&&&&1&&1&&&& \\
&&&0&&1&&2&&& \\
&&0&&0&&1&&3&& \\
&0&&0&&0&&1&&4& \\
0&&0&&0&&0&&1&&5
\end{array}$$

The sequence on the left-hand edge of this triangle is 0, 1, 0, 0, 0, 0, ...

3. (a) The boustrophedon array begins

$$\begin{array}{ccccccccccc}
&&&&&1&&&&& \\
&&&&1&&2&&&& \\
&&&4&&3&&1&&& \\
&&1&&5&&8&&9&& \\
&24&&23&&18&&10&&1& \\
1&&25&&48&&66&&76&&77
\end{array}$$

thus the boustrophedon transform—the terms at the opposite edge from where the original sequence was placed—is 1, 2, 4, 9, 24, 77, ...

(b) The boustrophedon array begins

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & -1 & & 0 & & & & \\
 & & & 0 & & 1 & & 1 & & & \\
 & & -1 & & -1 & & 0 & & 1 & & \\
 & 0 & & 1 & & 2 & & 2 & & 1 & \\
-1 & & -1 & & 0 & & 2 & & 4 & & 5
\end{array}
$$

thus the boustrophedon transform—the terms at the opposite edge from where the original sequence was placed—is 1, 0, 0, 1, 0, 5, ...

4. Similar to problem 2 the terms of the given sequence are placed alternately on the right- and left-hand edges of the triangle, in the positions where the boustrophedon transform of the desired sequence would be located, and then subtract to work backward to the desired sequence:

$$
\begin{array}{ccccccccccc}
 & & & & & 0 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 2 & & 1 & & 0 & & & \\
 & & 0 & & 2 & & 3 & & 3 & & \\
 & 4 & & 4 & & 2 & & -1 & & -4 & \\
0 & & 4 & & 8 & & 10 & & 9 & & 5
\end{array}
$$

The desired sequence is this 0, 1, 0, 0, -4, 0, ...

5. Emulate the recursive formula for the terms of the corresponding triangular array, noting that the odd-numbered rows are computed from left-to-right as in the triangular case, but the even-numbered rows are computed from right-to-left. Note that when listed, starting from the left end of a row, the entries of a triangular array are $S_{n,0}, S_{n,1}, S_{n,2}, \ldots$, while starting from the right end of a row they will be $S_{n,n}, S_{n,n-1}, S_{n,n-2}, \ldots$. This results in the recursive formula

$$
S_{n,k} = \begin{cases} a_n & n \text{ odd, } k = 0 \\ S_{n,k-1} + S_{n-1,k-1} & n \text{ odd, } k > 0 \\ a_n & n \text{ even, } k = n \\ S_{n,k+1} + S_{n-1,k} & n \text{ even, } k < n \end{cases}.
$$

6. It may seem natural to prove this using a double induction, on both n and k, or by using some form of strong induction that allows for assumptions on both induction variables, but in fact that is not necessary. The proof can be completed using ordinary induction on k only. This is because in the recursive definition of $T_{n,k}$ given in the background material, only value of T whose subscript is $k-1$ are needed.

Base case. For $k = 0$ and all values of n, the recursive formula is $T_{n,0} = a_n$ while the formula given in the problem evaluates to

$$\sum_{i=0}^{0} \binom{0}{i} a_{n-i} = \binom{0}{0} a_{n-0} = 1 \cdot a_n = a_n$$

so the formula holds true when $k = 0$.

Inductive step. Assume the formula holds for $k - 1$ (and all values of n). Then

$$\begin{aligned}
T_{n,k} &= T_{n,k-1} + T_{n-1,k-1} \\
&= \sum_{i=0}^{k-1} \binom{k-1}{i} a_{n-i} + \sum_{i=0}^{k-1} \binom{k-1}{i} a_{n-1-i} \\
&= \sum_{i=0}^{k-1} \binom{k-1}{i} a_{n-i} + \sum_{i=1}^{k} \binom{k-1}{i-1} a_{n-i} \\
&= \binom{k-1}{0} a_n + \sum_{i=1}^{k-1} \left(\binom{k-1}{i} + \binom{k-1}{i-1} \right) a_{n-i} + \binom{k-1}{k-1} a_{n-k}
\end{aligned}$$

where the third line is obtained by adjusting the index of the second sum, and the fourth by collecting the terms with the same index of a. A standard fact about binomial coefficients is $\binom{k-1}{i} + \binom{k-1}{i-1} = \binom{k}{i}$ (or this can be shown from the formula given in the question and properties of factorials). Also, the binomial coefficients at the extremities satisfy $\binom{k-1}{0} = \binom{k-1}{k-1} = 1 = \binom{k}{k} = \binom{k}{0}$. So the last line above simplifies to

$$T_{n,k} = \sum_{i=0}^{k} \binom{k}{i} a_{n-i}$$

which is the desired result.

7. The previous problem provides a direct formula for the triangular transform of a sequence $\{a_n\}$. Since the terms of the triangular transform appear at the right-hand ends of the rows of the corresponding triangular array, the formula is

$$b_n = T_{n,n} = \sum_{i=0}^{n} \binom{n}{i} a_{n-i}.$$

Since $\binom{n}{0} = 1$, this formula may be easily rearranged to

$$a_n = b_n - \sum_{i=1}^{n} \binom{n}{i} a_{n-i}.$$

This expresses a_n directly in terms of b_n and previous a's. Together with the initial condition $a_0 = b_0$, the result is a recursive formula for $\{a_n\}$:

$$a_n = \begin{cases} b_0 & n = 0 \\ b_n - \sum_{i=1}^{n} \binom{n}{i} a_{n-i} & n > 0 \end{cases}.$$

Alternately, the sequence $\{b_n\}$ can be placed on the right-hand edge of a triangular array, and terms to the left can be found by subtraction: $T_{n,k-1} = T_{n,k} - T_{n-1,k-1}$. This determines the unique value of $T_{n,k-1}$ that will provide the correct value of $T_{n,k}$. Together with the given values $T_{n,n} = b_n$ and a little induction this argument produces unique values of $a_n = T_{n,0}$ that will produce the sequence $\{b_n\}$.

8. In the result of the previous problem, note that $a_0 = b_0$ is an integer, and that each succeeding a is a sum of products of integers subtracted from an integer, leaving an integer. In the alternate solution, note that each entry of the triangle is found by subtracting an integer from an integer, so the inverse triangular transform, which are entries in the triangle, must be integers.

   (Technically an induction would be required here but for only one point the graders are only going to look for the argument about why the result is an integer, not the complete induction set up.)

9. Similar to the alternate explanation in problem 7, exploit the recursive definition of the boustrophedon array from problem 5 to recover the entire array from just the transformed sequence. Let $S_{0,0} = b_0$. Then assume all $S_{j,k}$ for $0 \leq k \leq j < n$ have been found. Proceed as follows:

   - If n is odd, let $S_{n,n} = b_n$. Then recursively define $S_{n,h-1} = S_{n,h} - S_{n-1,h-1}$ for all $n \geq h \geq 1$.
   - If n is even, let $S_{n,0} = b_n$. Then recursively define $S_{n,h} = S_{n,h-1} - Sn - 1, h - 1$ for all $1 \leq h \leq n$.
   - Take $a_n = S_{n,0}$ if n is odd and $a_n = S_{n,n}$ if n is even.

   Then, since subtraction uniquely inverts addition, these are the only possible value of $S_{n,k}$, and consequently a_n that allow $\{b_n\}$ to be the boustrophedon transform of $\{a_n\}$.

10. In the previous proof, note that subtraction of integers always leaves an integer so the entire array generated—including the a_n—will consist of integers.

11. $\langle\langle 1, 3, 2, 4\rangle\rangle, \langle\langle 1, 4, 2, 3\rangle\rangle, \langle\langle 2, 4, 1, 3\rangle\rangle, \langle\langle 2, 3, 1, 4\rangle\rangle, \langle\langle 3, 4, 1, 2\rangle\rangle$.

12. There are $\boxed{16}$ such permutations. This can be found by brute force, listing them all. A slick alternative to listing them all would be to note that the 5 must either be the second or fourth term, and each permutation where the 5 is second is matched by one where 5 is fourth by reversing the listing. If 5 is the second number, there are four choices for the first number, and two choices to arrange the last three numbers in up-down sequence, leading to $4 \times 2 = 8$ up-down permutations where 5 is the second number, and another 8 where it is the fourth number.

13. in the background material, a down-up permutation was obtained from an up-down permutation by reversing the order of the terms, but that trick only works for permutations with an even number of terms—the reverse of an odd-length up-down permutation is another up-down permutation. Instead, use the trick of turning the permutation

upside-down. If the permutation is of the first n positive integers, subtract each from $n+1$ to obtain another permutation of the first n integers, turning an up-down permutation into a down-up permutation and *vice versa*. For example, $\langle\langle 2,5,3,4,1\rangle\rangle$ becomes $\langle\langle 4,1,3,2,5\rangle\rangle$. Since this is a one-to-one correspondence between up-down and down-up permutations it demonstrates that there are the same number of each.

14. Since the number of up-down and down-up permutations for each n are the same, the number $2A_{n+1}$ counts the total number of alternating permutations of the first $n+1$ positive integers. The right-hand side of the recurrence relation can also be seen to count the total number of alternating permutations as follows.

    To find an alternating permutation of the first $n+1$ positive integers, first set aside the largest of them, $n+1$. For each $0 \le k \le n$ choose k of the integers from 1 to n. There are $\binom{n}{k}$ ways to do this. If k is even, arrange these integers in a down-up permutation (A_k ways to do this) and place them before $n+1$, and arrange the remaining integers in an up-down permutation (A_{n-k} ways to do this) and place them after $n+1$. If k is odd, arrange the first k integers in an up-down permutation instead.

    For example, if $n=6$ and $k=2$, choose two numbers from 1 to 6 (say, 3 and 5) and arrange them in a down-up (k is even) permutation. There is only one way to do this: $\langle\langle 5,3\rangle\rangle$. Now arrange the remaining four numbers (1, 2, 4, and 6) in an up-down permutation (say, $\langle\langle 2,4,1,6\rangle\rangle$). Stringing these choices together leads to the alternating permutation $\langle\langle 5,3,7,2,4,1,6\rangle\rangle$.

    All alternating permutations are accounted for in this manner, and for each choice k the number of alternating permutations where $n+1$ is in position $k+1$ is seen to be $\binom{n}{k}A_kA_{n-k}$. Summing for all possible k delivers the desired result.

15. To find A_6 set $n=5$ and compute

$$\begin{aligned} 2A_6 &= \sum_{k=0}^{5}\binom{5}{k}A_kA_{n-k} \\ &= \binom{5}{0}A_0A_5 + \binom{5}{1}A_1A_4 + \binom{5}{2}A_2A_3 + \binom{5}{3}A_3A_2 + \binom{5}{4}A_4A_1 + \binom{5}{5}A_5A_0 \end{aligned}$$

    Using the previously discovered values $A_0 = A_1 = A_2 = 1, A_3 = 2, A_4 = 5, A_5 = 16$ gives

$$2A_6 = 1\cdot 1\cdot 16 + 5\cdot 1\cdot 5 + 10\cdot 1\cdot 2 + 10\cdot 2\cdot 1 + 5\cdot 5\cdot 1 + 1\cdot 16\cdot 1$$

    which totals to 122, so $A_6 = 61$.

16. First, $E_{0,0} = S_{0,0} = 1$. If n is odd, $E_{n,0} = S_{n,0} = a_n = 0$ while if n is even $E_{n,0} = S_{n,n} = a_n = 0$ so the formula for $E_{n,k}$ is correct when $k=0$.

    When $n>0$ is odd, the definition gives $E_{n,k} = S_{n,k}$. For $k>0$, the solution to problem 5 gives $S_{n,k} = S_{n,k-1} + S_{n-1,k-1}$. But $n-1$ is even, so $S_{n-1,k-1} = E_{n-1,(n-1)-(k-1)} = E_{n-1,n-k}$ while $S_{n,k-1} = E_{n,k-1}$. Substituting, $E_{n,k} = E_{n,k-1} + E_{n-1,n-k}$, as desired.

    When $n>0$ is even, the definition gives $E_{n,k} = S_{n,n-k}$. For $k>0$, the solution to problem 5 now gives $S_{n,n-k} = S_{n,n-k+1} + S_{n-1,n-k}$. Since $n-1$ is odd, $S_{n-1,n-k} = E_{n-1,n-k}$

while $S_{n,n-k+1} = E_{n,n-(n-k+1)} = E_{n,k-1}$ so once again $E_{n,k} = E_{n,k-1} + E_{n-1,n-k}$, as desired.

17. When $k = 0$ the values are correct, for $E_{0,0} = 1$ and there is one down-up permutation of the first $0 + 1 = 1$ positive integers and it starts with 1, while if $n > 0$ there are no down-up permutations of $n + 1$ numbers starting with $k + 1 = 1$ because there are no positive integers less than one.

    Now assume the statement is has been shown for all $E_{m,h}$ whenever $m < n$ or $m = n$ and $h < k$. The number of down-up permutations of $n+1$ numbers starting with $k+1$ can be broken into two disjoint sets: those where the second number is exactly k and those where the second number is less than k.

    If the second number is less than k then another down-up permutation of the first $n+1$ positive integers can be obtained by swapping the first number, $k+1$, for k, whatever position it is in. For example, $\langle\langle 4, 2, 3, 1, 5\rangle\rangle$ can have $k + 1 = 4$ swap positions with $k = 3$ to obtain $\langle\langle 3, 2, 4, 1, 5\rangle\rangle$. This operation can be reversed, so it is a one-to-one correspondence between down-up permutations of $n + 1$ numbers that start with $k + 1$ whose second number is *not* k, and down-up permutations of $n+1$ numbers with start with k. By the inductive assumption, this latter is equal to $E_{n,k-1}$.

    If the first number in the permutation is $k + 1$ and the second number is exactly k, a permutation of the first n positive integers can be obtained by deleting the first number and subtracting one from all other entries that are larger than k. For example, $\langle\langle 3, 2, 5, 1, 4\rangle\rangle$ can be changed into $\langle\langle 2, 4, 1, 3\rangle\rangle$. This transformation can be reversed, so it is a one-to-one correspondence between down-up permutations of $n+1$ numbers that start with $k + 1$ whose second number is k, and *up-down* permutations of n numbers that start with k. By the correspondence in problem 13, this is the same as the number of down-up permutations of n numbers that start with $n + 1 - k = (n - k) + 1$, so by the inductive assumption these permutations are counted by $E_{n-1,n-k}$.

    Thus, the total number of down-up permutations of $n + 1$ numbers starting with $k + 1$ is $E_{n,k-1} + E_{n-1,n-k} = E_{n,k}$.

18. From the previous problem, $E_{n,n}$ counts the number of down-up permutations of the first $n + 1$ positive integers that start with $n + 1$. Deleting the first number, the remaining terms of the permutation form an up-down permutation of the first n positive integers, which is the definition of A_n.